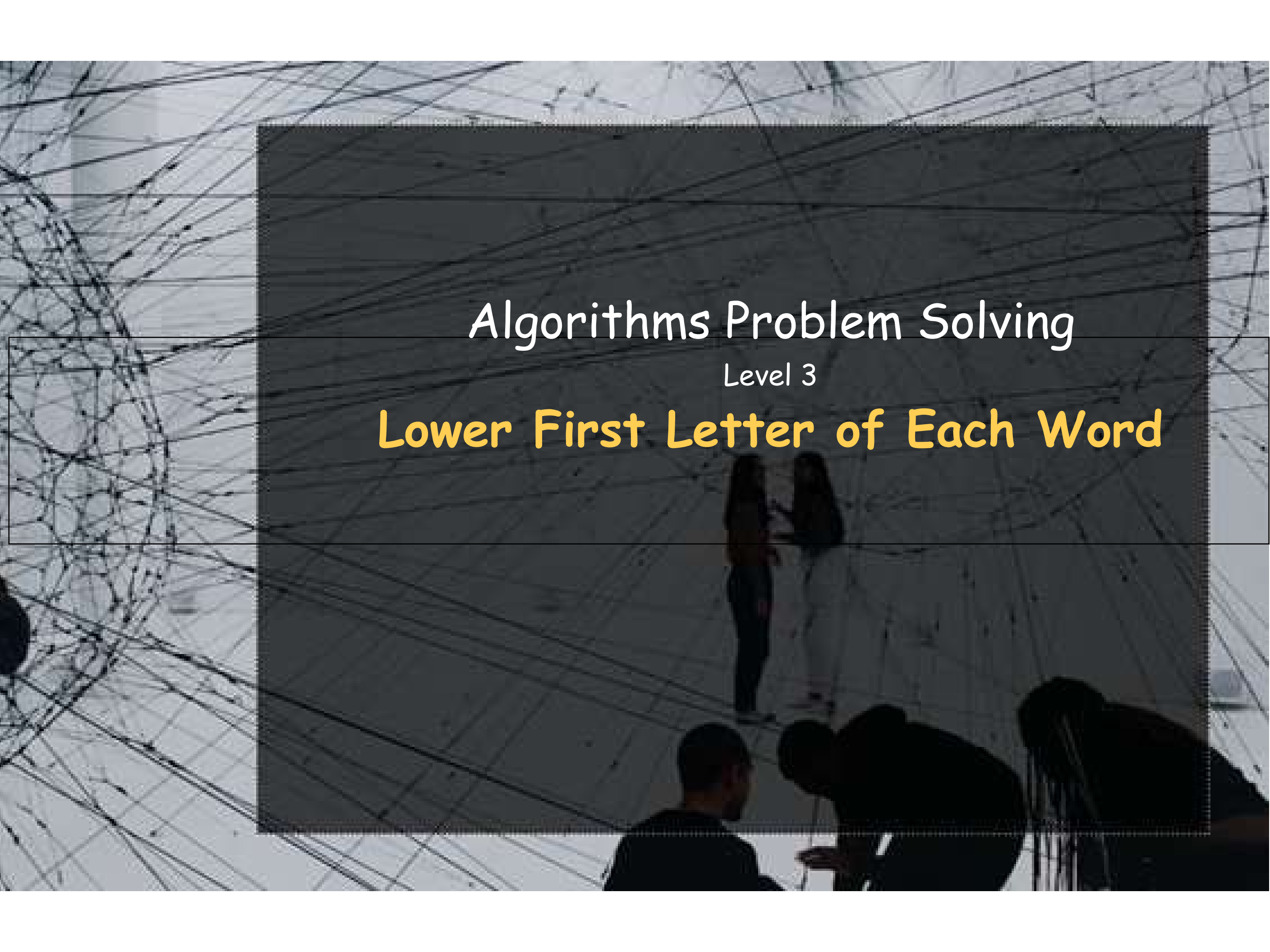


Algorithms Problem Solving

Level 3

Lower First Letter of Each
Word



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Definition

Word-Initial Lowercase Transformation

-  Program converts the first letter of every word to lowercase.
-  The remaining characters retain their existing case.

Explanation

Word Boundary Detection

 The program identifies the beginning of each word by checking the beginning of the string or a preceding whitespace character.

Lowercase Conversion

 The detected first letter is passed to ``tolower()``.

 Only the target character is modified.

Remaining Character Preservation

 Characters that are not word initials are left unchanged.

 This makes the function different from converting the entire string to lowercase.

Procedures

- 1 Read the text string.
- 2 Detect each word start.
- 3 Apply ``tolower()`` to the first letter.
- 4 Preserve all remaining characters.
- 5 Display the modified string.

💡 Example

Input:

Sara Alterkaw

➡ Result

sara alterkawi

Note

- ◆ The operation changes only word-initial characters.
- ◆ The rest of each word keeps its original case.
- ◆ The function can be useful when transforming text into certain programming-style formats.

⊘ Warning

- ◆ Do not confuse this function with `LowerAllString()`.
- ◆ Punctuation and symbols may affect how a word boundary is detected.
- ◆ Use `tolower()` carefully with character types from `<cctype>`.



Comparison

▨ `LowerFirstLetterOfEachWord()`

- ✓ Changes only the first character of each word.
- ✓ Preserves the case of remaining characters.
- ✓ Performs targeted lowercase transformation.

▤ `LowerAllString()`

- ✓ Converts every applicable character to lowercase.
- ✓ Changes the entire string.
- ✓ Does not preserve existing uppercase letters.

✦ Summary

- ✓ Detects the beginning of each word.
- ✓ Converts only the leading character to lowercase.
- ✓ Preserves the remaining character casing.
- ✓ Demonstrates selective string transformation.

Thank you 😊